Chapter 25

PALAEOMAGNETISM OF THE BALTIC SHIELD — IMPLICATIONS FOR PRECAMBRIAN TECTONICS

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ABSTRACT

Palaeomagnetic data of the Precambrian (age 2700—800 Ma) rocks from the Baltic (Fennoscandian) Shield are reviewed. In the light of new palaeomagnetic pole positions and radiometric age data, a revised apparent polar wander path (APWP) is presented for the Baltic Shield. Apart from the Sveconorwegian (1000—800 Ma ago) results there are no major discrepancies in the pole positions of widely separated rock units, suggesting that the Baltic Shield has behaved as a coherent unit since early Precambrian times. The shape of the APW path of the Baltic Shield differs from that of the Laurentian Shield, indicating independent drift of the shields at least during part of the Precambrian. The dissimilar palaeolatitude curves of the two shields support this idea. However, during the Svecokarelian/Hudsonian orogeny (age 1900 Ma) and during the Jotnian/MacKenzie magmatic interval (1300—1200 Ma ago), the two shields had similar palaeolatitudes and may have been in juxtaposition. A collision of the two shields of these times could have triggered the Svecokarelian/Hudsonian orogenies 1900 Ma ago and the global rifting and magmatic episode (Jotnian/MacKenzie) about 1250 Ma ago. An intimate connection between the Shields was evident also during the Grenville/Sveconorwegian orogeny. This is suggested by palaeomagnetic and geological data.

The APW speed (c. 0.35°/Ma) of the Baltic Shield during the Precambrian is significantly lower than that of the Laurentian Shield (0.50°/Ma). Remanent magnetization with "normal" polarity is dominant in the Baltic Shield.

INTRODUCTION

Palaeomagnetic data of Precambrian rocks can be used to analyze whether pre-Mesozoic horizontal motions have taken place within or between different shields. Relative movements between lithospheric units are indicated by differences in the APW paths of the units. If the different provinces of a shield show different APW paths, the plate tectonic model can be used to describe the evolution of the shield (Piper et al., 1973; Cavanaugh and Seyfert, 1977). However, if the APW curves are similar, the ensialic within-shield tectonics is the proper model (McElhinny and McWilliams, 1977). Dissimilar APW curves of different shields indicate that the shields have drifted independently and emphasize that the opening and closing of oceans is a recurrent feature in the earth's history (Wilson, 1966; Spall, 1973).

In order to find out whether relative motions of lithospheric units have

taken place, accurate APW paths are needed for each shield. In this paper we present a new APW path for the Baltic Shield which supersedes the simpler paths previously proposed (e.g. Neuvonen, 1970; Spall, 1973; Poorter, 1975). The new APW curve is based on all the available palaeomagnetic data of the Baltic and Ukrainian Shields. The data include more than ten new palaeopole determinations, all of which have radiometric age control (Neuvonen et al., 1981; Pesonen et al., in prep.; Pesonen and Suominen, in prep.; Pesonen, in prep.; O. Kouvo, pers. comm., 1980). The APW curve, which covers the age interval 2700—800 Ma, is compared with that of the Laurentian Shield. Finally, the palaeolatitude curves of the Baltic and Laurentian Shields are compared and their relevance to Precambrian tectonics is discussed.

GEOLOGICAL SETTING

The Baltic Shield (Fig. 25-1) can be subdivided into five different blocks: (1) the Pre-Karelian basement, (2) the Granulite Complex of Finnish Lapland, (3) The Svecokarelian belt, (4) the Sveconorwegian province and (5) the Caledonian belt (Phanerozoic) which is not discussed in this paper (see Poorter, this volume, Chapter 24).

The palaeomagnetic data are arranged into groups according to the provinces mentioned. In addition, two major post-orogenic or anorogenic events involving igneous activity are recognized within the Svecokarelian belt: the sub-Jotnian (1650—1350 Ma ago) and the Jotnian (1300—1200 Ma ago). These magmatic intervals are treated as individual groups, because palaeopoles of these rocks are of crucial importance in defining the APWP of the Baltic Shield. The sampling areas are shown in Fig. 1 and are summarized in Table 25-I.

The pre-Karelian (early Precambrian) basement in eastern Finland, Sweden and the northwestern USSR is composed of granites, gneisses and supracrustal rocks with U-Pb ages generally older than 2500 Ma (Kouvo, 1976; Gaál et al., 1978). The rocks of this area were, however, greatly affected by deformation and metamorphism during the Svecokarelian orogeny (1900 Ma ago). The early Karelian rocks (age 2400—2000 Ma), which overlie the basement in its marginal area, are grouped together with the pre-Karelian data. Thirteen palaeopoles belong to this group (Table 25-I).

The infracrustal rocks of the Granulite Complex form an arcuate zone in northern Finland (Fig. 25-1). This province has suffered low- to high-grade metamorphism several times, as shown by radiometric ages which appear to peak around 2500 Ma, 2100 Ma and 1900 Ma (Meriläinen, 1976). It is palaeomagnetically interesting that the last major metamorphism about 1900 Ma ago coincides with the Svecokarelian orogeny. Two palaeomagnetic poles are available from this province (Fig. 25-1; Table 25-I).

The Svecokarelian belt consists of various types of infracrustal and

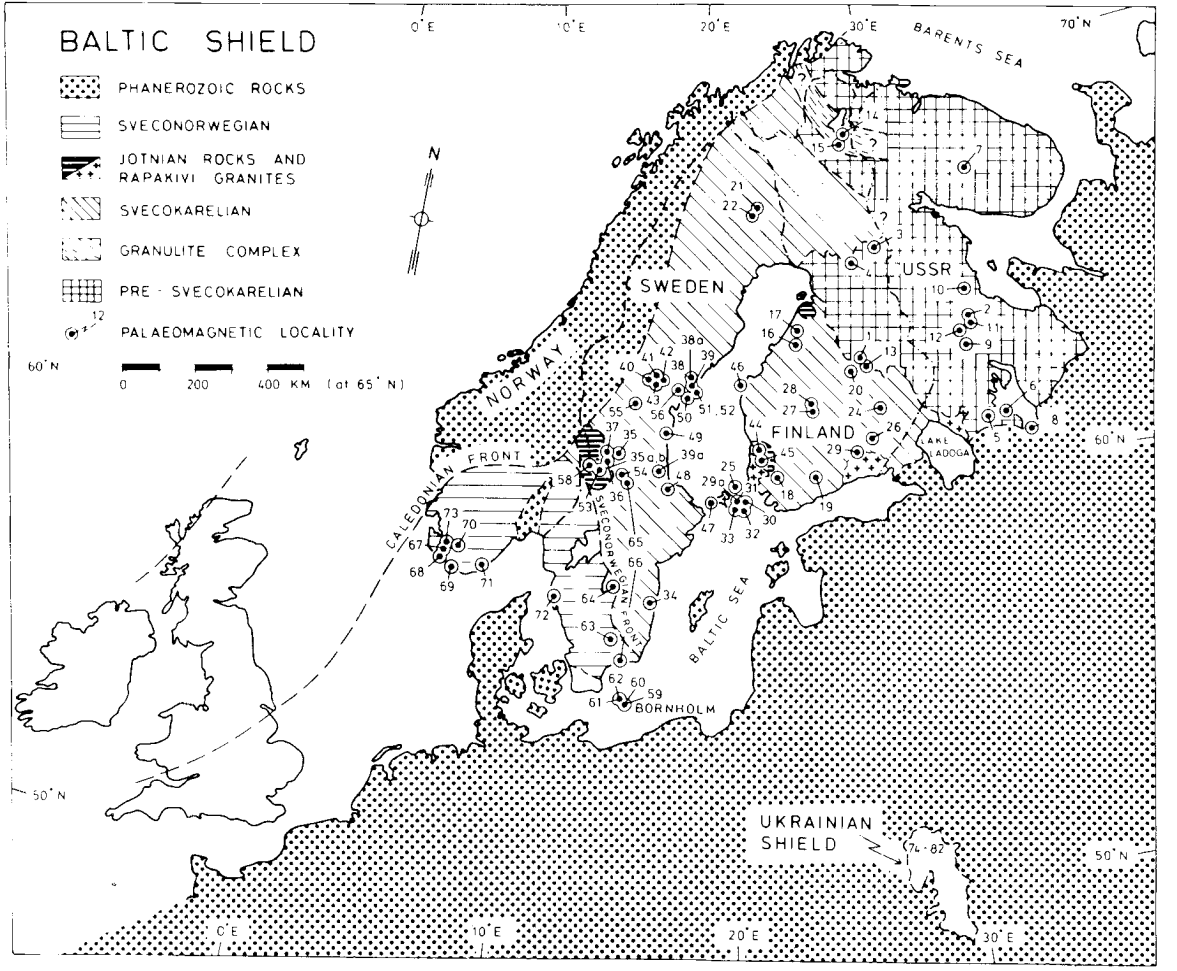


Fig. 25-I. Simplified geologic map of the Baltic Shield showing the five main provinces and the magmatic episodes. The continuation of the Granulite Complex across the Finnish border is not known is detail. The pre-Svecokarelian basement also includes both older (age > 2900 Ma) and younger (presumably Jatulian or Svecokarelian) inliers which are not shown. Palaeomagnetic sampling sites are marked with numbered circles (see Table 25-I).

Symbol	Rock unit	Country	S/B/N	Age	Method	Palaeomagnetic pole Lat. (N), Long. (E)	dp, dm (A_{95})	Pol.	Cat.	Refs./Notes
I. Pre-Svecokarelian										
1	Nilsiä diorites	F	1/8/64	2684	c	63.9, 313.0	7, 8	N	A	(1), (2)
2	Jatulian beds and lavas	SU	1/12/38	1870—1610		58.0, 238.0	6, 12	N	A	(3)
3	Jatulian dikes and greenstones	F	1/4/47	2150	c	47.2, 233.4	11, 18	N	A	(4), (5)/b
4	Syöte gabbro	F	1/1/11	2440	c	35.7, 221.2	2, 4	N	B	(4), (6)/b
5	Pukhta—Pedaselsk group	SU	1/-/57	1950—1850	a, d	44.0, 231.0	3, 5	N, R	A	(7)
6	Shoksha group	SU	1/-/29	1950—1850	a, d	54.0, 240.0	3, 5	N	A	(7)
7	Kola gabbro-norites	SU	1/-/12	1880—1820		40.0, 223.0	9, 16	N	A	(7)
8	Onega gabbro-diabases	SU	1/-/15	1800—1600		32.0, 230.0	1, 2	N	A	(7)
9	Central Karelia dikes	SU	1/2/12	1870—1610		49.0, 235.0	5, 7	N	A	(7)
10	Northern Karelia sandstones	SU	1/2/7	1870—1610		65.0, 217.0	8, 10	N	B	(7)
11	Central Karelia sandstones	SU	1/5/5	—		57.0, 239.0	15, 21	N	B	(7)
12	Central Karelia sandstones	SU	1/3/14	—		62.0, 264.0	5, 7	N	B	(7)
13	Varpaisjärvi dikes	F	1/11/17	—		47.0, 188.0	4, 6	N	B	(1), (2)
II. Granulite Complex										
14	Akujärvi quartz diorite	F	1/3/10	1925	c	41.3, 245.5	4, 7	N	A	(8), (9)
15	Laanila dikes	F	1/3/18	—		− 3.5, 218.1	5, 8	N	B	(8)
III. Svecokarelian										
16	Ylivieska gabbro	F	1/5/42	1885	c	43.3, 242.4	5, 8	N	A	(10)
17	Pohjanmaa intrusives	F	1/5/45	~ 1900		37.9, 239.1	8, 15	N	B	(10)
18	South Finland intrusives	F	1/7/82	~ 1900		39.7, 242.6	10, 18	N	B	(11)
19	Hyvinkää gabbro	F	1/11/11	1875	c	44.1, 246.2	5, 9	N	A	(12), (13)
20	Pielavesi gabbro	F	1/5/20	~ 1900		35.9, 238.2	2, 4	N	B	(8), (14)
21	Tärendö gabbro	S	1/5/17	2000—1800		44.6, 228.7	10, 17	N	A	(15)
22	Tärendö silicic rocks	S	1/3/12	2000—1800		42.0, 248.0	—	N	B	(15)
23	Mean synorogenic intrusives	F, S	7/41/229	~ 1900		41.2, 240.8	(4.3)	N	A	--/(d)
24	Rantasalmi lamprophyres	F	1/11/24	1837	c	49.0, 226.0	2, 3	N	A	(1), (2)
25	Åva intrusives	F	1/15/-	1815—1803	c	41.0, 169.0	9, 17	N	B	(16)
26	Luontarivesi granodiorite	F	1/25/211	1790—1765	c	67.0, 198.0	10, 17	N	A	(17), (18)
27	Keuruu dikes	F	1/11/34	~ 1880	c	45.4, 218.0	4, 7	N	A	(19), (2)/(c)
28	Keuruu dikes	F	1/3/12	⩽ 1900		26.3, 257.8	2, 3	R	B	(19), (2)/(c)
Sub-Jotnian igneous interval (~ 1650—1320 Ma)										
29	Kuisaari dolerite	F	1/8/84	1650	c, e	22.0, 190.0	7, 14	R	A	(17), (20)
29a	Åland rapakivi	F	1/1/6	1659—1589	c	19.5, 194.4	15, 18	N	B	(21), (20)
30	Kumlinge—Brändö dikes	F	1/7/25	1602—1556	b	12.2, 182.0	5, 9	R	A	(22)
31	Föglö—Sottunga dikes	F	1/6/25	1523—1518	b, c	27.9, 187.5	6, 13	N	A	(22)
32	Kumlinge dikes	F	1/4/17	1602—1556	b	14.3, 190.1	8, 14	R	A	(16)/(b)
33	Föglö dikes	F	1/8/56	1523—1518	b, c	31.3, 186.5	6, 12	N	A	(23)
34	Loftahammar gabbro	S	1/5/26	1694—1460	b	23.0, 179.0	2, 5	N	A	(24), (25)
35	Dala porphyries	S	1/2/11	806— 701/1570	a/b	23.0, 184.0	—	N	A	(26), (27)
35a	Dala metasediments	S	1/6/27	—		33.0, 177.0	7, 14	M	B	(28)
35b	Dala volcanics	S	1/8/46	—		27.0, 189.0	5, 11	M	B	(28)
36	Öje basalts	S	1/2/11	931— 745/1405	a/b	32.2, 186.0	5, 10	N	A	(26), (27)

Table 25-I (Continued)

Symbol	Rock unit	Country	S/B/N	Age	Methods	Palaeomagnetic pole Lat. (N), Long. (E)	dp, dm (A_{95})	Pol.	Cat.	Refs. Notes
37	Dala dolerites	S	1/4/19	—		23.0, 178.0	21, 40	N	B	(26), (27)
38	Nordingrå gabbro-anorthosite	S	1/7/39	⩾1385	e	34.1, 135.7	8, 14	R	A	(28)/(b)
38a	Nordingra gabbro-anorthosite	S	1/12/69	⩾1385		38.5, 141.6	4, 7	N	A	(28)/(b)
39	Nordingra granite	S	1/1/28	1415		33.0, 149.0	8, 14	R	A	(28)/(b)
39a	Gävle granite	S	1/1/12	—		35.0, 165.0	7, 14	N	B	(28)
40	Ragunda intrusive rocks	S	1/11/-	~1320		51.3, 169.5	7, 11	N	A	(25)/(b)
41	Ragunda intrusive rocks	S	1/4/-	~1320		53.2, 163.7	15, 23	R	B	(25)/(b)
42	Ragunda dolerites	S	1/3/-	1320—1250		9.2, 195.1	19, 34	N	B	(25)/(b)
43	Ragunda dolerites	S	1/9/-	1320—1250		16.2, 194.2	10, 20	R	B	(25)/(b)
Jotnian igneous interval (~ 1300—1150 Ma)										
44	Satakunta sandstone	S	1/10/40	~1300	a	3.0, 180.0	8, 12	N	B	(29), (30)
45	Satakunta dolerites	F	1/18/62	1236—1225	b	2.0, 158.0	3, 5	N	A	(31), (32)
				1263	c					
46	Vaasa dolerite dikes	F	1/15/58	1225/1270	b/c	7.0, 164.0	3, 6	N	A	(33), (32)
47	Märket dolerite dikes	F	1/8/58	1270	b, c	— 5.9, 145.5	7, 11	N	A	(23), (32)
48	Gävle dolerites	S	1/2/15	1245		8.0, 150.0	2, 4	N	A	(24), (35)
49	Gnarp dolerites	S	1/1/6	1245	b	—11.0, 159.0	7. 10	N	B	(24), (35)
50	Nordingra dolerites	S	1/2/12	1245	b	— 5.0, 158.0	5, 9	N	A	(24), (35)
50a	Nordingrå baked rocks	S	1/13/80	1250	e	— 2.3, 157.4	3, 5	N	A	(28)
51	Ulvö dolerites	S	1/-/32	1245	a	1.0, 161.0	5, 9	N	A	(36)
52	Ulvö dolerites	S	1/-/20	1245	a	— 4.0, 153.0	4, 6	N	A	(37)
53	Älvdalsasen dolerite sills	S	—	1231	b	— 1.4, 148.6	5, 6	N	B	(35)
54	Älvho dolerite sills	S	—	1290—1215	b	17.3, 170.2	9, 13	N	B	(35)
55	Sundsjö—Gimän dolerite dikes	S	—	1229—1156	b	2.9, 147.4	6, 11	N	B	(35)
56	Väst-Norrland dolerites	S	1/43/252	1245	a	— 7.5, 156.5	2, 3	N	A	(38)
57	Mean Jotnian dolerites	S, F	13/-/-	~1250		1.0, 156.0	(5.3)	N	A	—/(d)
58	Dala dolerites and basalts	S	1/1/5	~1200		—10.0, 115.0	15, 27	N	B	(39)
59	Vasea dike	DB	1/1/11	~1200		—16.0, 127.8	6, 10	N	B	(40)/(a)
60	Listed dike	DB	1/1/32	~1000		13.8, 250.1	2, 4	N	B	(40)/(a)
61	Bolshavn dike	DB	1/1/6	~1000		20.7, 223.6	7, 14	N	B	(40)/(a)
62	Kjeldsea dike	DB	1/3/11	~1100—1000		7.0, 179.0	6, 10	R	B	(41), (40)/(a)
IV. Sveconorwegian (~ 1100—800 Ma)										
63	Swedish hyperites	S	1/2/10	1516— 886	a	— 6.0, 240.0	(18.0)	R	A	(26), (27)/(b)
64	Swedish hyperites	S	1/3/60	1573— 781	a	—19.8, 213.3	(24.1)	R	A	(26), (27)/(b)
65	Falun—Karlshamn dolerites	S	1/4/28	987— 890	b	— 9.7, 233.4	(32.1)	R	A	(42)/(b)
66	Falun—Karlshamn dolerites	S	1/4/24	1005— 890	b	13.7, 243.8	(15.1)	N	A	(42)/(b)
67	Mean Egersund dolerites	N	2/10/88	880— 663	a, e	—25.0, 231.5	(13.3)	R	A	(43), (44)/(d)
68	Mean Egersund anörthosites	N	2/9/-	~1000— 850	a	—39.6, 203.6	(16.1)	N	A	(45), (46)/(d)
69	Egersund farsundites	N	1/7/35	920— 900	a	—43.3, 194.0	9, 10	N	A	(47)
70	Hunnedalen dikes	N	1/8/37	⩽950— 850	a, b	—34.0, 208.0	9, 10	N	A	(43)/(d)
71	Bamble—Kongsberg basement	N	1/6/48	1120— 975	a, b, c	— 8.0, 208.0	6, 8	N, R	B	(34)/(d)
72	Tuve dolerite	S	1/1/10	—		—17.0, 238.9	2, 3	R	B	(48)
73	Mean Rogaland basement	N	1/3/19	950— 850	a, b	—36.0, 233.0	10, 12	N	A	(43)/(d)
V. Ukrainian Shield										
74	Bug—Bodolsky complex	UR	—	2300—2000	a, b, c	54.0, 286.0		R	B	(49), (50), (51)

Table 25-I (Continued)

Symbol	Rock unit	Country	S/B/N	Age	Methods	Palaeomagnetic pole Lat. (N), Long. (E)	dp, dm (A_{95})	Pol.	Cat.	Refs. Notes
75	Ingulets complex	UR	—	2000—1700	a, b, c	64.0, 325.0		R	B	(49), (50), (51)
76	Ingulets complex	UR	—	2000—1700	a, b, c	57.0, 316.0		R	B	(49), (50), (51)
77	Bokovyansk intrusion	UR	—	1700—1500	a, b, c	52.0, 301.0		R	B	(49), (50), (51)
78	Takova granite	UR	—	1600—1400		51.0, 326.0		R	B	(49), (50), (51)
79	Turchinki gabbro	UR	—	1400—1200		12.0, 212.0		R	B	(49), (50), (51)
80	Uman granite	UR	—	1750	a, b, c	26.0, 240.0		R	B	(49), (50), (51)
81	Korosten complex	UR	—	1750	a, b, c	24.0, 232.0		R	B	(49), (50), (51)
82	Korosten complex	UR	—	1750	a, b, c	27.0, 238.0		R	B	(49), (50), (51)

Symbols	refer to Figs. 25-1 and 25-2.
Country	DB = Bornholm (Denmark); F = Finland; N = Norway; S = Sweden; SU = Soviet Union; UR = Ukraine.
S/B/N	number of studies/sites/samples.
Age/Method	radiometric ages (Ma) recalculated, when possible, with the new decay constants (Steiger and Jäger, 1977). a = K-Ar; b = Rb-Sr; c = U-Pb; d = U-Pb-Th; e = geological evidence. For further details of the ages see references listed.
Lat. (N), Long. (E)	*latitude and longitude (degrees) of the palaeomagnetic pole.*
dp, dm	95% confidence oval of the pole (single study).
(A_{95})	95% confidence circle of the mean pole (several studies combined).
Pol.	polarity of the pole: N = normal, R = reversed (see p. 629).
Cat.	Category of the pole. A means reliable data, B less reliable in a manner outlined on p. 631.
Notes	(a) for other Bornholm poles and their interpretation models, see Abrahamsen (1977); (b) normal and reversed poles recalculated by the present authors from original data; (c) see text; (d) some recalculations performed in this study.

References: (1) *Neuvonen et al. (1981);* (2) *O. Kouvo (pers. commun., 1980);* (3) Irving and Hastie (1975); (4) Neuvonen (1975); (5) Sakko (1971), (6) Kouvo (1977); (7) McElhinny and Cowley (1977); (8) Pesonen (in prep.); (9) Meriläinen (1976); (10) Pesonen and Stigzelius (1972); (11) Neuvonen (1974); (12) R. P. Puranen (pers. commun., 1980); (13) Härme (1978); (14) Helovuori (1979); (15) Cornwell (1968); (16) Neuvonen (1970); (17) Neuvonen (1978); (18) Korsman and Lehijärvi (1973); (19) Pesonen et al. (in prep.); (20) Vaasjoki (1977); (21) Johnson (1979); (22) Pesonen and Suominen (in prep.); (23) Neuvonen and Grundström (1969); (24) Poorter (1976a); (25) Piper (1979b); (26) Mulder (1971); (27) Priem et al. (1968); (28) Piper (1980); (29) Neuvonen (1974); (30) Simonen (1980); (31) Neuvonen (1965); (32) Suominen (in prep.); (33) Neuvonen (1966); (34) Poorter (1975); (35) Patchett et al. (1978); (36) Magnusson and Larson (1977); (37) Larson and Magnusson (1976); (38) Piper (1979a); (39) Dyrelius (1970); (40) Abrahamsen (1977); (41) Schønemann (1972); (42) Patchett and Bylund (1977); (43) Poorter (1972); (44) Storetvedt and Gidskehaug (1968); (45) Hargraves and Fish (1972); (46) Murthy and Deutsch (1974); (47) Murthy and Deutsch (1975); (48) Abrahamsen (1974); (49) Kruglyakova (1961); (50) Spall (1973); (51) Semenenko et al. (1968).

supracrustal rocks which were affected by the major Svecokarelian orogeny about 1900 Ma ago (Kouvo, 1976). Thirteen palaeopoles have been obtained from this belt. The sampling localities extend from northern Sweden to southeastern Finland (Fig. 25-1).

The sub-Jotnian palaeomagnetic data consist of 20 palaeopoles located within the Svecokarelian belt in Finland and Sweden. Similarly, 14 palaeopoles are available from the Jotnian sedimentary and igneous rocks of the belt in Sweden and Finland (Table 25-I).

The youngest Precambrian tectonic province in the Baltic Shield is the Sveconorwegian block for which radiometric ages range from 1100 Ma to 800 Ma (Versteeve, 1975; Lundqvist, 1979), although older relict ages are also reported (Skiöld, 1976). These ages, especially those determined by the K-Ar method, most likely reflect the cooling and uplift of the Sveconorwegian province. The province is surrounded in the west by the Caledonian belt and in the east by the Sveconorwegian front, which appears to be a 50—100 km broad, subvertical tectonic boundary (the schistosity zone; Fig. 25-1; e.g. Patchett and Bylund, 1977). The similarity in geology and radiometric age data of the Sveconorwegian and Grenville provinces (Laurentian Shield) is striking and has been noted by several workers (e.g. Wynne-Edwards and Hasan, 1970; Zwart and Dornsiepen, 1978; *see also Baer, this volume, Chapter 14, ed.*). There are eleven palaeopole measurements from this province, including those from the schistosity zone and adjacent dyke swarms (the Falun-Karlshamn dykes).

In addition to the Baltic Shield, Precambrian bedrock is exposed as numerous minor shield inliers in Europe (e.g. Zwart and Dornsiepen, 1978). In this paper we have included the palaeomagnetic data only from the Ukrainian Shield (Fig. 25-1) from which nine poles are reported by Kruglyakova (1961; Table 25-I).

THE APWP OF THE BALTIC SHIELD

Table 25-I lists all the palaeomagnetic poles from the Baltic and Ukrainian Shields in the age interval 2700—800 Ma. Methods and results of radiometric dating are also given along with references to original studies.

The polarity has been defined so that north-seeking magnetization, which yields palaeomagnetic poles in the Pacific region (Fig. 25-2), is called "normal". The poles that would plot on the opposite hemisphere are called "reversed" and are inverted by 180 degrees before plotting. The true polarity of Precambrian poles is, however, uncertain because there are large gaps in the APW paths (Fig. 25-2), and segments of polar wander paths cannot always be reliably connected (Spall, 1973; Irving and McGlynn, 1976). Moreover, the "normal" polarity with respect to the Baltic and Laurentian Shields may differ.

In this paper we distinguish between the two possible polarities (although

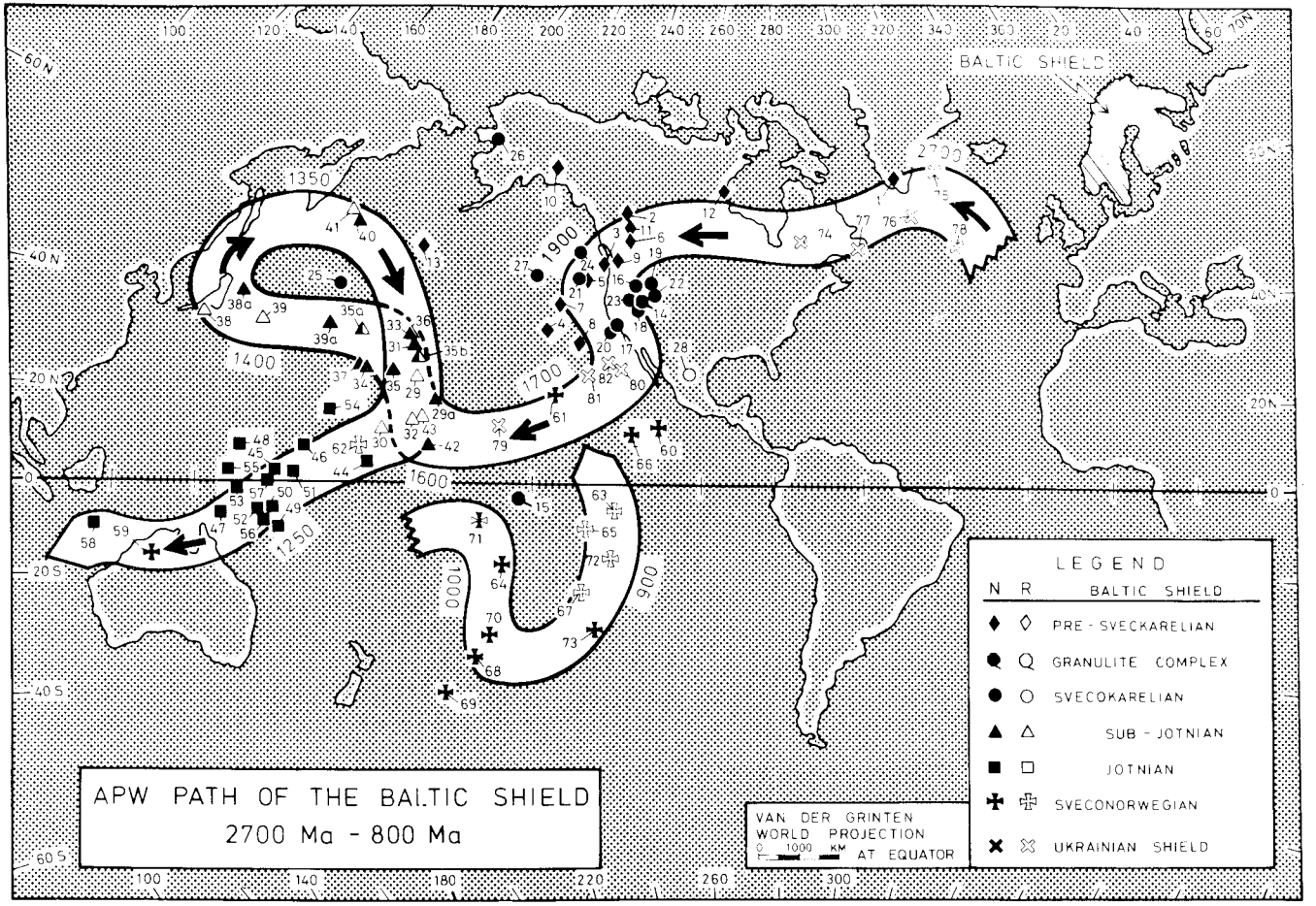


Fig. 25-2. The apparent polar wander path (APWP) for the Baltic Shield (2700—800 Ma ago). The pole positions have the same numbers as the sampling areas of Fig. 25-1. The 95% confidence limits for the poles are given in Table 25-I. Open (closed) symbols denote reversed (normal) polarity. For other symbols see legend.

this involved some recalculation), because systematic departures from the 180° symmetry of antipoles may render important clues as to unremoved secondary components, non-dipole disturbances or apparent polar wander during the polarity change (see detailed discussion in Pesonen, 1978).

Each pole in Table 25-I is characterized by a reliability category symbol, A or B. Category A includes the more reliable data for which a radiometric age is available and reported demagnetization studies show no indication of unremoved secondary components (see also Irving and Hastie, 1975; Irving and McGlynn, 1976). The majority (53%) of the data belong to category A. The remaining data (47%) are less reliable and comprise category B. A few results (about 5%), which are claimed to be unreliable by the original authors (e.g. Neuvonen, 1967), were excluded from this study.

GENERAL DESCRIPTION OF THE APWP

The simplest APW path for the Baltic Shield is a smooth swathe (Piper et al., 1973) with a width of 15° (Fig. 25-2).

The APW swathe starts from pole 1 of probable Archaean age and located in Greenland (Fig. 25-2), passes through the North American continent, makes a large hairpin turn in the Pacific (the sub-Jotnian loop) 1650—1350 Ma ago before arriving in northern Australia (poles 58—59) at the end of the Jotnian interval (1200 Ma ago). Then there is a major gap in palaeomagnetic data as no poles are available for the age interval 1200—1000 Ma. Therefore, the Sveconorwegian poles (63—73) are treated as a separate track representing the age interval 1000—800 Ma (Fig. 25-2).

The main differences between the APW path shown in Fig. 2 and the previously proposed paths are the new early Precambrian—Svecokarelian track (2700—2000 Ma ago) and the large northerly sub-Jotnian loop (1650—1350 Ma ago) (Pesonen, 1979b; Piper, 1980).

Pre/early Karelian data (2700—1900 Ma)

The only early Precambrian pole comes from dioritic rocks in the Nilsiä region of east-central Finland (Figs. 25-1 and 25-2), for which radiometric dating yields an age of 2680 Ma (Neuvonen et al., 1981). Fold or baked contact tests have not been conducted, but other evidence indicates an early Precambrian age for the remanent magnetization. The pole position differs by more than 60° from the early Karelian (2—13) and synorogenic Svecokarelian poles (16—28) with ages between 2400 Ma and 1800 Ma (Pesonen and Stigzelius, 1972; Neuvonen, 1974) but some of the Ukrainian poles (74—78) which appear to have either Archaean or early Karelian ages (2400—1700 Ma) plot in this same area (see also Semenenko et al., 1968; Spall, 1973).

The early Karelian palaeomagnetic poles (age 2400—1900 Ma) are located

on the western coast of North America (poles 2, 3, 5, 6, 9 and 11). These poles appear to form a cluster which, in the APW swathe, slightly predates the subsequent Svecokarelian (1900 Ma) poles (16—24). The pre- and early Karelian poles are of "normal" polarity except for the Pukhta-Pedaselsk group (pole 5), which displays short "reversed" intervals (Katseblin, 1968). The Ukrainian poles (74—78) are all of "reversed" polarity (Fig. 25-2).

Early Karelian poles deviating from the APW swathe also occur. Poles 4, 7 and 8 (Fig. 25-2) show small deviations which may be explained by differences in the radiometric and palaeomagnetic uplift ages (e.g. Pullaiah et al., 1975; Poorter, this volume, Chapter 24). Poles 10 and 13 depart significantly from the APW path, as discussed later (p. 633).

Granulite Complex data

Two poles (14, 15) are available from the Granulite Complex (Figs. 25-1 and 25-2). The first one is obtained from the Akujärvi quartz diorite, for which radiometric dating yields an U-Pb age of 1925 Ma (Meriläinen, 1976). This remanent magnetism with normal polarity was probably recorded during the uplift and slow cooling of the quartz diorite, which took place after granulite facies metamorphism about 2000—1900 Ma ago (Pesonen, in prep.). The other pole from the Granulite Complex is derived from a swarm of olivine diabase dykes (pole 15; Pesonen, in prep.) which cut the granulites in the Laanila region. No radiometric age data are available for these dykes. The pole position (Fig. 25-2) suggests either an age of about 1700 Ma or about 900 Ma. Both of these ages are realistic on geological grounds. Diabase dykes with an age of 1730 Ma exist in the Lake Inari area about 100 km north of Laanila (Meriläinen, 1976), and igneous activity at about 1150—900 Ma is known in northern Finland (R. Lauerma, pers. commun., 1980), in Sweden (Kresten et al., 1977) and in the Kola Peninsula (T. Mutanen, pers. commun., 1980).

Svecokarelian palaeomagnetic data

In defining the APW path for the Baltic Shield, the pole for the synorogenic Svecokarelian intrusives (age 1900—1800 Ma; Neuvonen, 1974) is of crucial importance. The mean pole (23) is determined from seven rock units in Sweden and Finland. Consistency between the Swedish (21, 22) and Finnish (16—20) poles is excellent (Fig. 25-2).

Poles 27 and 28 are also of considerable importance in defining and interpreting the APW path for the Baltic Shield. These poles come from the Keuruu diabase dykes (age 1880 Ma; Pesonen et al., in prep.) in central Finland (Fig. 25-1 and 25-2). The dykes must have been intruded into

an already reasonably cold crust as they have chilled margins against the synorogenic Keuruu gabbro. Furthermore, both normal and reversed polarities are present in the dykes (Pesonen et al., in prep.) in contrast with the synorogenic rocks, which are all of normal polarity (Fig. 25-2; Table 25-I). The reversal in the Keuruu dykes is asymmetric (not 180°). Baked contact tests and detailed demagnetization studies suggest that the asymmetry is not caused by secondary remanence. For this reason the poles (27, 29) have not been averaged. The APW interpretation is, however, difficult as age data are available only for a normally magnetized dyke. We will return to the problem of asymmetric reversals on p. 637.

Poles 24—26 are significantly younger (age about 1815 Ma) than the synorogenic intrusives (age about 1900 Ma; Table 25-I). The pole positions also differ from those of the synorogenic intrusives (Fig. 25-3). Accordingly, we consider that a prominent anti-clockwise polar loop is possible during the age interval 1850—1750 Ma as shown in Fig. 25-3. Because it is based on only four poles, this post-orogenic APW loop must be regarded as tentative.

Sub-Jotnian loop

A gap of almost 60° exists in the APW path in the age interval 1750—1650 Ma.

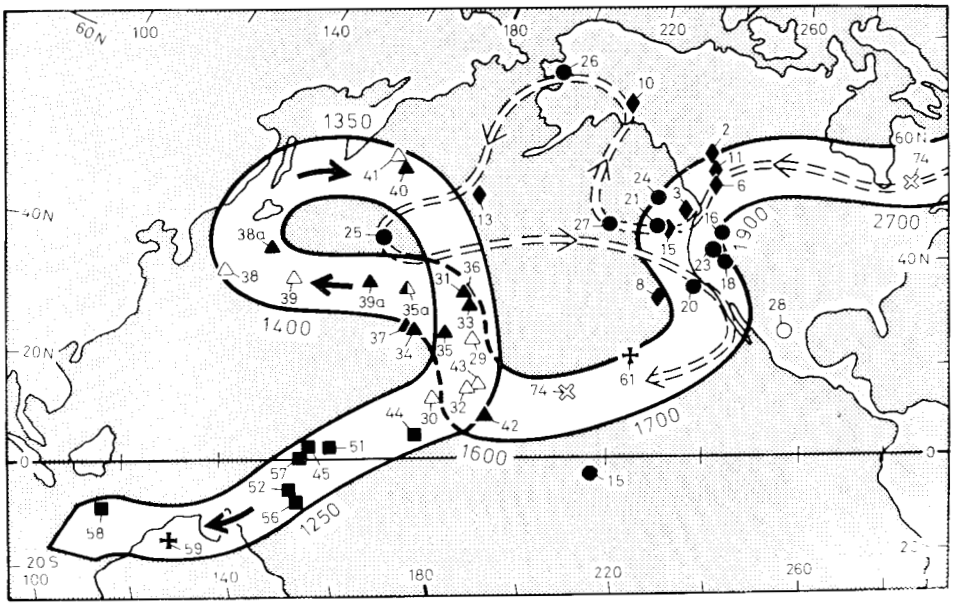


Fig. 25-3. An alternative APW path for the Baltic Shield. A more complex path is shown as a dashed line superimposed on the APW path of Fig. 25-2. Symbols as in Fig. 25-2.

Only a group of Ukrainian poles (80—82) and possibly poles 15 and 61 fit this part of the APW track (Fig. 2).

The oldest sub-Jotnian results come from the Kuisaari dolerite (29) (Neuvonen, 1978), the Åland Rapakivi massif (29a) (Johnson, 1979) and the Kumlinge dykes in the Åland archipelago (30, 32) (Neuvonen and Grundström, 1969; Pesonen and Suominen, in prep.). The Kuisaari dolerite was magnetized at the time of Ahvenisto rapakivi intrusion (1650 Ma ago; Vaasjoki, 1977), as verified by positive baked contact tests (Neuvonen, 1978). The Kumlinge dykes were magnetized during the intrusion of the dykes, which took place somewhat later (about 1600 Ma ago) than the emplacement of the Åland rapakivi massif (about 1650 Ma ago; Vaasjoki, 1977; Suominen, in prep.). This is supported by baked contact tests and by the reversed polarity of the dykes in contrast with the normal polarity of the Åland rapakivi massif (Fig. 25-2; Table 25-I; Pesonen and Suominen, in prep.).

Poles 33—37 form a tight cluster and represent the age interval of about 1570—1500 Ma. They are all of normal polarity (Neuvonen and Grundström, 1969; Mulder, 1971; Poorter, 1976a; Piper, 1980; Pesonen and Suominen, in prep.). In this group the Swedish poles (34—37) agree well with the Finnish poles (31, 33), demonstrating internal consistency of the Baltic Shield palaeomagnetic data. The depth of the sub-Jotnian loop is established by the poles of the Gävle granite (pole 39a), the Nordingrå intrusions (poles 38, 38a and 39; ages about 1415—1385 Ma) and the Ragunda intrusion (age about 1320 Ma) (poles 40, 41; Piper, 1979b, 1980). The return part of the loop is defined by poles 42 and 43 of the post-Ragunda dolerites (age $\leqslant$ 1320 Ma), which cut the Ragunda intrusion (Piper, 1979b).

Jotnian results

The palaeomagnetic poles (45—47) of the Jotnian magmatic episode (age 1300—1200 Ma) form a cluster situated slightly to the northeast of Australia (Fig. 25-2; Neuvonen, 1965, 1966, 1970; Neuvonen and Grundström, 1969; Poorter, 1976a; Larson and Magnusson, 1976; Magnusson and Larson, 1977; Patchett et al., 1978; Piper, 1979a, b, 1980). The poles of the Ragunda dykes (age $\leqslant$ 1320 Ma), Jotnian sandstone (age 1300 Ma) and Jotnian dyke rocks and sills (age 1250 Ma) all lie in the APW swathe in accordance with geological and geochronological data (Table 25-I). Poles 58 (Dalå dolerites; Dyrelius, 1970) and 59 (Vaseå dyke; Abrahamsen, 1977) deviate from the other Jotnian poles to the younger side on the APW swathe. This deviation, associated with category B poles, could result from causes other than apparent polar wander. The deviation is, however, in accordance with the age (1200 Ma) of these poles which suggests them to be younger than the rest of the Jotnian poles (Fig. 25-2). The mean pole of the Swedish Jotnian dolerites (Lat. = $-0.2°$ N, Long. = $156.0°$ E, $A_{95} = 6.3°$) is in excellent agreement

with the corresponding Finnish pole (Lat. = 5.0° N, Long. = 155.8° E, A_{95} = 5.0°). All the Jotnian poles are of "normal" polarity, consistent with the corresponding data from MacKenzie intrusive rocks of North America (Patchett et al., 1978).

The second major hiatus in the APW swathe is the age interval 1200—1000 Ma. Therefore, the path between the Jotnian poles and the Sveconorwegian poles is left open (e.g. Abrahamsen, 1977).

The Sveconorwegian loop

The oldest Sveconorwegian poles are derived from highly metamorphosed basement rocks (71, 73; Poorter, 1972, 1975) and from synorogenic anorthosite intrusives (68, 69; Hargraves and Fish, 1972; Murthy and Deutsch, 1974, 1975). These rocks have been magnetized during their slow cooling and uplift after the Sveconorwegian metamorphism. Their magnetization age can be only roughly fixed between the time of metamorphism and final uplift to the approximate time interval 1000—900 Ma ago.

Problems arise not only in age definitions but also in interpretation of complex magnetizations. Multicomponent remanence is indicated by the nonlinear vector diagrams and the asymmetric reversals in these rocks (Mulder, 1971; Poorter, 1972, 1975).

The younger arm of the Sveconorwegian loop is composed of poles from several dyke swarms, some of which are located in the Sveconorwegian province (Hunnedalen (70), Tuve (72) and Egersund (67) dolerites; Storetvedt and Gidskehaug, 1968; Poorter, 1972; Abrahamsen, 1974). Others are situated either in the schistosity zone (Swedish hyperites (63—64; Mulder, 1971) or in the adjacent terrain (Falun-Karlshamn dykes (65—66), Patchett et al., 1978).

The Hunnedalen dykes probably record an uplift magnetization as their pole position is very close to that of the anorthosites (68) and farsundites (69; Fig. 25-2). The Egersund and Tuve dolerites record intrusion magnetization. The Egersund dykes are clearly younger (880—663 Ma) than the basement (ages 1000—900 Ma; Table 25-I). The dykes also have chilled margins and opposite polarity compared with the basement rocks (Storetvedt and Gidskehaug, 1968).

The complex magnetization history of the Swedish hyperites (63—64) is manifested by their multicomponent remanence (Mulder, 1971; Poorter, 1972). Results of baked contact tests are not available, and it is not known whether the magnetization reflects a Sveconorwegian overprint or the true intrusion age. The presence of two polarites (with strong asymmetry) supports the latter idea but the wide K-Ar age range (1573—781 Ma; Priem et al., 1968) favours the former alternative.

Positive results were obtained from baked contact tests performed on radiometric and palaeomagnetic data of the Karlshamn-Falun dykes (Patchett

et al., 1978). This suggests that the dykes were magnetized during their intrusion into a cold Svecokarelian terrain just outside the schistosity zone. These dykes give the first palaeopoles of Sveconorwegian age (1000—800 Ma), obtained from rocks outside the Sveconorwegian block. The reversal in these dykes is asymmetric, a feature which appears to be characteristic of the Sveconorwegian reversals (Fig. 25-2).

Four poles (59—62) are available from the dolerite dykes of the Bornholm island (Schønemann, 1972; Abrahamsen, 1977). Their magnetism is either of Jotnian (poles 59, 62) or of Sveconorwegian (poles 60, 61) age. The geology of Bornholm is, however, very complex as the island lies close to the Sveconorwegian front and to the Polish Trough (Fig. 25-1)(Abrahamsen, 1977; Pozaryski and Brochwicz-Lewinski, 1978). Hence, the Bornholm poles have been included in category B.

PROBLEMS IN THE BALTIC SHIELD APWP

Before we attempt to interpret the palaeomagnetic results of the Baltic Shield in terms of various tectonic models, it is worthwhile to consider causes other than plate motion which can affect the polar paths of Figs. 25-2 and 25-3. Errors in sampling and measurement, insufficiently eliminated secular variation and discrepancies between radiometric and magnetization ages can cause considerable scatter of palaeopoles (*see also Irving and McGlynn, this volume, Chapter 23, and McWilliams, this volume, Chapter 26, ed.*).

Secondary components

The Precambrian rocks of the Baltic Shield have been affected by several metamorphic events (e.g. Kouvo, 1976; Meriläinen, 1976; Gaál et al., 1978), each of which has added another component to the magnetic and radiometric record carried by the rocks. A single metamorphic event involving total remagnetization permits the treatment of the new characteristic remanence by standard methods, and the resulting radiometric and palaeomagnetic ages relate only to the times of metamorphism. In the case of partial remagnetization the new NRM consists of at least two components. Recent palaeomagnetic studies in North America (e.g. Buchan and Dunlop, 1976; McGlynn and Irving, 1978) demonstrate that multicomponent NRMs are common in Precambrian metamorphic rocks. The secondary components can be isolated by modern demagnetization techniques (Roy and Lapointe, 1978), vector subtraction (Buchan and Dunlop, 1976) great circle methods (Halls, 1976), fully executed baked contact tests and palaeointensity measurements (Pesonen, 1978, 1979a). Such investigations of multicomponent NRMs have considerably modified the APWP of North America (Fig. 25-5).

There are no detailed studies on rocks from the Baltic Shield where

secondary components have been properly separated. Yet in many cases the existence of secondary remanence is clearly indicated by non-linear vector diagrams (Mulder, 1971; Poorter, 1972, 1975) or asymmetric reversals (e.g. Mulder, 1971; Patchett et al., 1978; Pesonen et al., in prep.). The secondary overprints due to Svecokarelian, Sveconorwegian or Caledonian orogenies should be interesting subjects for future studies.

Priem et al. (1968) suggested that the sub-Jotnian palaeopoles represent remagnetization due to the Caledonian orogeny because these poles lie very close to the Palaeozoic poles of Europe. The Caledonian remagnetization hypothesis can, however, be rejected (Neuvonen, 1970; Spall, 1973) on the following grounds. The poles from the oldest (2700—1900 Ma) to the youngest (1300—1200 Ma) rocks (Fig. 25-2) form a chronologic sequence which continues through and beyond the Palaeozoic poles. It also seems unlikely that all the diverse rock types, representing widely separated areas and different metamorphic grades, would be completely remagnetized. Further, although there is a wide range of K-Ar ages, none of them is Palaeozoic. Finally, the baked contact test for both normal and reversed sub-Jotnian dykes indicates remanence acquisition during intrusion and not during a regional remagnetization (Pesonen and Suominen, in prep.).

The internal consistency of the palaeomagnetic data of various ages in the Baltic Shield (e.g. the Svecokarelian, sub-Jotnian and Jotnian rocks), the positive results of baked contact tests (e.g. Patchett et al., 1978; Neuvonen et al., 1981; Pesonen et al., in prep.) and the discovery of new field reversals (Piper, 1979b, 1980; Pesonen et al., in prep.) make it unlikely that the smooth chronological sequence of poles in Fig. 25-2 has been caused by remagnetization. Future studies should, however, be carried out to clarify whether the scatter of poles within the swathes and the differences between the swathes of Figs. 25-2 and 25-3 can be explained by unremoved secondary components.

Asymmetric reversals

It is noteworthy that almost all of the palaeomagnetic reversals observed in the Baltic Shield show departures from the 180° symmetry (e.g. reversals 27—28, 30—31, 63—64 and 65—66). In some cases this may be caused by unremoved secondary components, but the baked contact tests, geochemical data and radiometric ages suggest that the asymmetry is not always due to secondary overprinting (Pesonen, in prep.). The asymmetry can also be caused by non-dipolar geomagnetic field disturbances, short period geomagnetic excursions or apparent polar wander (e.g. plate motion) during the polarity change (e.g. Pesonen, 1979a). The last possibility is interesting as it would mean that relatively detailed APWP portions can be established. The portions connecting two poles of opposite polarity can be fixed if the

relative ages of the two poles can be determined by geological means (see detailed discussion of this topic in Pesonen, 1978).

TECTONIC IMPLICATIONS OF THE APWP

From the discussions presented above we may conclude that the APWP is produced by tectonic causes such as motion of the Baltic Shield as a whole, differential uplift within the shield (Ueno and Irving, 1976; Neuvonen, 1978) or relative motion of individual parts of the shield. The last two mechanisms are of special interest regarding the evolution of the shield itself, depending on whether ensialic within-shield tectonics or Wilson-cycle plate tectonics dominated (McElhinny and McWilliams, 1977; *see also McWilliams, this volume, Chapter 26, ed.*).

Differential uplift and APWP

Morgan (1976) and Beckmann et al. (1977) have shown that smooth APW tracks can be produced if the uplift after regional metamorphism takes place differentially, so that one block rises more rapidly than nearby blocks (see also Ueno and Irving, 1976). In principle, the entire Precambrian APW path could be caused by differential uplift of subunits of the shield. For example, a suitable combination of regional tilt and uplift gives rise to a smooth polar wander path (Morgan, 1976).

Neuvonen (1978) tested this model in the Baltic Shield from the APW data with ages between 1800 Ma and 1600 Ma. He showed that the model cannot explain the APW track. Neuvonen based his arguments on the internal consistency of the Svecokarelian poles (25 and 26) and of the sub-Jotnian poles (29 and 32), which were derived from widely separated rock units (Fig. 25-1). This result does not preclude the possibility that small, especially post-metamorphic parts of the APW paths were produced by differential uplift.

Tectonics within the Baltic Shield

The internal consistency of the Svecokarelian (poles 21—22 and 16—20), sub-Jotnian (34—37 and 31—33) and Jotnian poles (poles 48—56 and 45—47) in Sweden and Finland suggests that major tectonic rotations or relative displacements have not taken place in this province since the Svecokarelian orogeny.

The Sveconorwegian poles of Sweden and Norway (e.g. poles 67 and 72) also show good consistency, but the areal distribution of sampling sites is restricted (Fig. 25-1). The consistency of palaeomagnetic poles from the Archaean and the Granulite Complex cannot be determined from the scarce data available.

Comparison of palaeomagnetic poles of rocks with similar age but from different tectonic blocks of the Baltic Shield can only be made in a few cases because of lack of data.

The 2400—1900 Ma old poles derived from different inliers of the early Precambrian province (e.g. poles 3—4 vs. poles 5—12; Fig. 25-1) all plot in the same region (Fig. 25-2). Similarly, the poles of the Akujärvi quartz-diorite from the Granulite Complex (pole 14) and the synorogenic intrusives from the Svecokarelian province (poles 16—22) lie close to one another although these provinces are separated by large Archaean rock sequences. These examples suggest that no major movements have taken place between the tectonic blocks of the Baltic Shield. Therefore, it seems that within-shield tectonics has dominated during the evolution of the Baltic Shield since pre-Svecokarelian times. The tectonic style during the Archaean (before 2400 Ma) cannot yet be determined on palaeomagnetic grounds as there is only one early Precambrian pole available.

Ukrainian Shield vs. Baltic Shield

The palaeomagnetic poles from the Ukrainian Shield (Kruglyakova, 1961) are plotted in Fig. 25-2 together with the poles from the Baltic Shield. It is evident that the Ukrainian poles from two groups display a reasonable fit with the APW curve of the Baltic Shield (see also Spall, 1973).

The first group formed by poles 74—78 fits into the early Precambrian—Svecokarelian APW track (Fig. 25-2) although the radiometric ages with a wide range (about 2400—1700 Ma; Semenenko et al., 1968) suggest that they may be younger. The other group of poles (80—82) is characterized by an age of 1750 Ma (Table 25-I), which is in accordance with the Baltic Shield palaeomagnetic data. Further, the poles and ages (79) of the Turchingi gabbro (1400 Ma) and Ragunda dykes (42—43; $\leqslant$ 1320 Ma) are rather similar. Although the demagnetization studies of Ukrainian rocks are limited, we can tentatively conclude that the Ukrainian results fit well with the APW curve of the Baltic Shield. This suggests that no major rotations have taken place between these two European Shields since Archaean time.

Baltic Shield vs. Laurentian Shield

The APW path of the Baltic Shield is compared to that of the Laurentian Shield in Fig. 25-4. The APW curve of the Laurentian Shield is compiled from the APW paths determined by Irving and McGlynn (1976) and McElhinny and McWilliams (1977), except for the 1350—1000 Ma interval which is taken from Pesonen (1978). The comparison to follow is restricted only to the data from North American cratons. Piper (1980) has made a similar comparison including the data from Greenland and Scotland. Several major differences between the two APW paths are recognized (Fig. 25-4). The

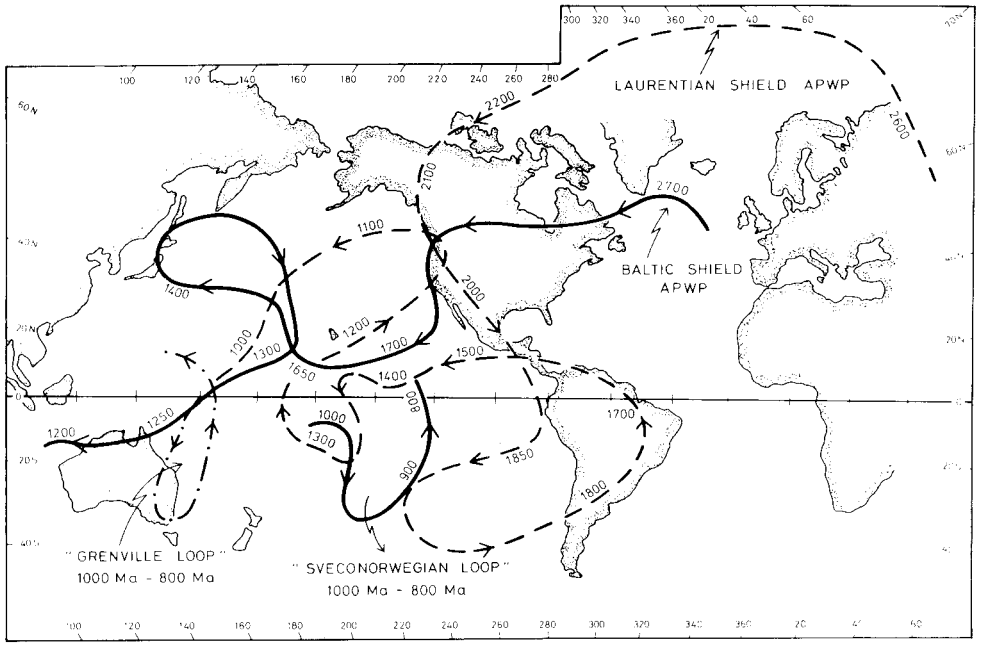


Fig. 25-4. Comparison of the APW paths of the Baltic and Laurentian Shields. The APW path of Fig. 25-2 is used for the Baltic Shield. The Laurentian APW path is compiled from data by Irving and McGlynn (1976; age interval 2200—800 Ma), McElhinny and McWilliams (1977; age interval 2200—2700 Ma) and Pesonen (1979a; age interval 1300—1000 Ma). Both curves are shown in present geographic coordinates. Closing the Atlantic Ocean into its pre-Mesozoic state (Bullard et al., 1965) would move the Baltic Shield poles approximately 38° to the west.

overall shapes of the two paths are clearly different. For example, the curve for the Baltic Shield shows a prominent sub-Jotnian polar wander loop during the age interval 1650—1320 Ma. In the Laurentian APW path a corresponding loop is missing, although recent palaeomagnetic results from Greenland (Piper, 1980) give indications of a somewhat similar loop. However, there is a major anticlockwise loop (Logan loop) in the Laurentian path during the age interval 1200—1000 Ma (Pesonen, 1979a). Such a loop has not yet been detected in the Baltic APW path. This may, however, simply result from the lack of palaeomagnetic data for this age interval. If the APW path of Fig. 25-3 is adopted for the Baltic Shield, there exists an additional anticlockwise loop at about 1850—1750 Ma ago. A somewhat similar loop is recognized in the Laurentian APW path (1850—1600 Ma) but the shapes of the two loops are different (Fig. 25-4). Moreover, as shown by McElhinny and McWilliams (1977), the Laurentian loop of this age interval may extend much more to the east than shown in Fig. 25-4.

Some similarities in the two APW paths are also observed. For example,

the early Precambrian—Svecokarelian/Hudsonian tracks (2700—1900 Ma) have similar shapes, although the Laurentian track of this age interval is much longer. The only tracks which are strikingly similar in both curves are the Sveconorwegian and Grenville loops of the age interval 1000—800 Ma. In both cases the poles form a hairpin-shaped, anticlockwise loop in the Southern Pacific region. Moreover, the closing of the Atlantic Ocean (Bullard et al., 1965) would cause the two loops to coincide (see also Donaldson et al., 1973; Buchan, 1978). Thus the palaeomagnetic data of the Baltic Shield support the idea of contiguity of the Grenville and Sveconorwegian provinces during the Sveconorwegian/Grenville orogenies (Poorter, 1976b; Patchett and Bylund, 1977). Similarities in geology (Wynne-Edwards and Hasan, 1970) and geochronological data (Patchet and Bylund, 1977) also support this conclusion (*see also Baer, this volume, Chapter 14, ed.*).

Recent palaeomagnetic results from the Laurentian Shield (Morris and Roy, 1977) appear to have resolved the "Grenville problem" and showed that the Grenville (about 1000—800 Ma old) and the Keweenawan (1200—1000 Ma old) tracks can be linked together without plate-tectonic suture mechanisms.

The corresponding problem in the Baltic Shield is the tectonic relation between the Sveconorwegian province and the rest of the shield (Fig. 25-2). This problem awaits solution as there are as yet no palaeomagnetic data for the age interval 1200—1000 Ma.

COMPARISON OF PALAEOLATITUDES

The palaeolatitude curves of the Baltic and Laurentian Shields are compared in Fig. 25-5. Both palaeolatitude curves have been derived directly from the APW paths of Fig. 25-4 using the method of Irving (1964, p. 186). The Baltic Shield curve was calculated with respect to the city of Kajaani which is located in the central part of the Shield (Fig. 25-1). The reference city for the Laurentian Shield is Winnipeg (see also Donaldson et al., 1973). In places where gaps exist in the APW curves, interpolations were made assuming constant APW speed. Although this method to produce the palaeolatitude curves has its limitations, the validity of the method was verified by comparing the palaeolatitude curve with actual palaeolatitude data (Baltic Shield) calculated directly from the category A palaeomagnetic results. This comparison (Fig. 25-5) shows that the data agree well with the smooth palaeolatitudinal curve.

The overall shapes of the two palaeolatitude curves are not similar. As the APW curves were also different, it seems probable that the shields have drifted independently during most of the Precambrian (Spall, 1973; Neuvonen, 1974). This conclusion is also supported by the different APW speeds of the shields. The velocity derived from the Baltic Shield curve (Fig. 25-2 and 25.3) is approximately 0.35°/Ma (range 0.31—0.40°/Ma), depending on

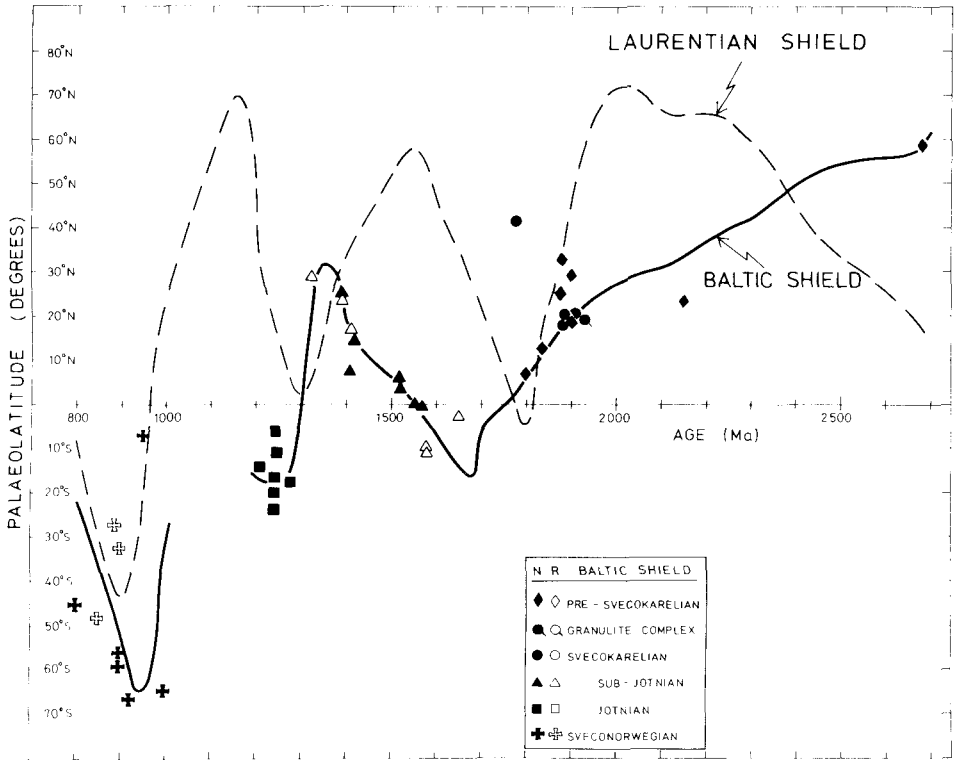


Fig. 25-5. Palarolatitude curves for the Baltic and Laurentian Shields. See p. 641 for explanation. The reference cities for the palaeolatitudes are Kajaani (Lat. = 64°13′, Long. 27°44′E; Baltic Shield) and Winnipeg (Lat. = 49°53′, Long. 97°10′W; Laurentian Shield).

whether the curve of Fig. 25-2 or 25-3 is used. The velocity of about 0.5°/Ma (Fig. 25-5) obtained for the Laurentian Shield is significantly higher.

According to Fig. 25-5 the two shields may have been in juxtaposition during three periods. The first period, indicated by crossing palaeolatitudes, is located in the age interval 1900—1750 Ma. Therefore, the cause of the Svecokarelian and coeval Hudsonian orogenies may have been a collision of the two shields about 1900 Ma ago. After this period the palaeolatitude curves depart and meet again about 1350—1250 Ma ago (Fig. 25-5). This second crossing of the latitudes may have some relation to the worldwide rifting and magmatic activity during the age interval 1300—1200 Ma (Jotnian/MacKenzie episodes; Neuvonen, 1974; Sawkins, 1976). The intimate connection of the Laurentian and Baltic Shields during this time is supported by the similarities in geology and magnetic properties ("normal" polarities) of the Jotnian/MacKenzie dykes (Patchett et al., 1978).

The palaeoclimatological conditions revealed by the metasediments on

both shields agree well with the palaeolatitudes calculated and consequently support the idea of mutual collisions at about 1900 Ma and 1300 Ma ago (Neuvonen, 1974). Similar geological features in addition to the palaeomagnetic data support the contiguity of the Grenville and Sveconorwegian provinces during the Grenville/Sveconorwegian deformation.

CONCLUSIONS

The following conclusions can be drawn from the presently available Precambrian palaeomagnetic data of the Baltic Shield.

(1) The systematic distribution of palaeopoles along a smooth APW swathe in chronologic sequence between 2700 Ma and 800 Ma ago is due to a motion of the Baltic Shield relative to the pole. Sampling and measurement errors, unremoved secondary components and differences between radiometric and magnetization ages explain the scatter of poles within the APW path.

(2) The palaeomagnetic data for rocks with the same age but from different tectonic provinces are consistent. This suggests that the Baltic Shield has behaved as a coherent unit at least since late Archaean times. Further, the consistency indicates that ensialic tectonics has been dominant during the evolution of this shield. The Sveconorwegian poles differ significantly from those of the other provinces of the Baltic Shield. Whether this reflects suturing between the Sveconorwegian and other blocks of the shield can be tested only after palaeomagnetic data are obtained for the age interval 1200—1000 Ma.

(3) The APW curves of the Baltic and Laurentian Shields are not similar, suggesting that both shields have drifted independently during the Precambrian. However, the palaeolatitude data indicate that the shields may have been in juxtaposition about 1900 Ma, 1350—1250 Ma and 1000—800 Ma ago. Collision of the two shields 1900 Ma ago could have initiated the major orogenies of Svecokarelian and Hudsonian times (1900—1800 Ma ago). The crossing palaeolatitude curves of the two shields 1300 Ma ago may bear some relationships to the worldwide rifting and igneous activity (Jotnian/MacKenzie interval) about 1250 Ma ago. The contiguity during the Grenville/Sveconorwegian deformation (1000—800 Ma ago) is supported by the similarity of the two areas.

(4) The APW velocity of the Baltic Shield (0.35°/Ma) is significantly lower than that (0.50°/Ma) of the Laurentian Shield. In order to verify this difference, additional data are needed for the Baltic Shield, where "normal" polarity data appear to dominate.

(5) The palaeomagnetic data from the Ukrainian Shield are consistent with the results from the Baltic Shield. This suggests that no major relative movements have taken place tween these two European shields since the Archaean (*see also Poorter, this volume, Chapter 24, ed.*).

(6) Future studies of secondary components, asymmetric reversals and scatter of palaeopoles should be carried out on the basis of detailed demagnetization experiments. Further palaeomagnetic work should be focused on rocks in the age interval 1200—1000 Ma in order to solve the problem whether the Sveconorwegian province has always been part of the Baltic Shield (ensialic tectonics) or whether it has collided and sutured with the rest of the shield (plate tectonics).

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